Weather Data Analysis

Comprehensive Meteorological Analysis Report

Project 3: Weather Domain Analysis

Report Date: September 08, 2026

Dataset: 365 Days of Weather Data (2023)

Executive Summary

This report analyzes weather observations over 365 days to identify temperature patterns, rainfall distribution, seasonal variation, weather conditions, and extreme observations.

KEY FINDINGS:
- Annual Temperature Average: 25.03°C
- Annual Rainfall: 2,447.97mm
- Rainy Days: 284 out of 365 (78% of year)
- Heavy Rain Days: 77, concentrated mainly during June-September
- Weather Pattern: Strong seasonal rainfall concentration consistent with a monsoon-like pattern

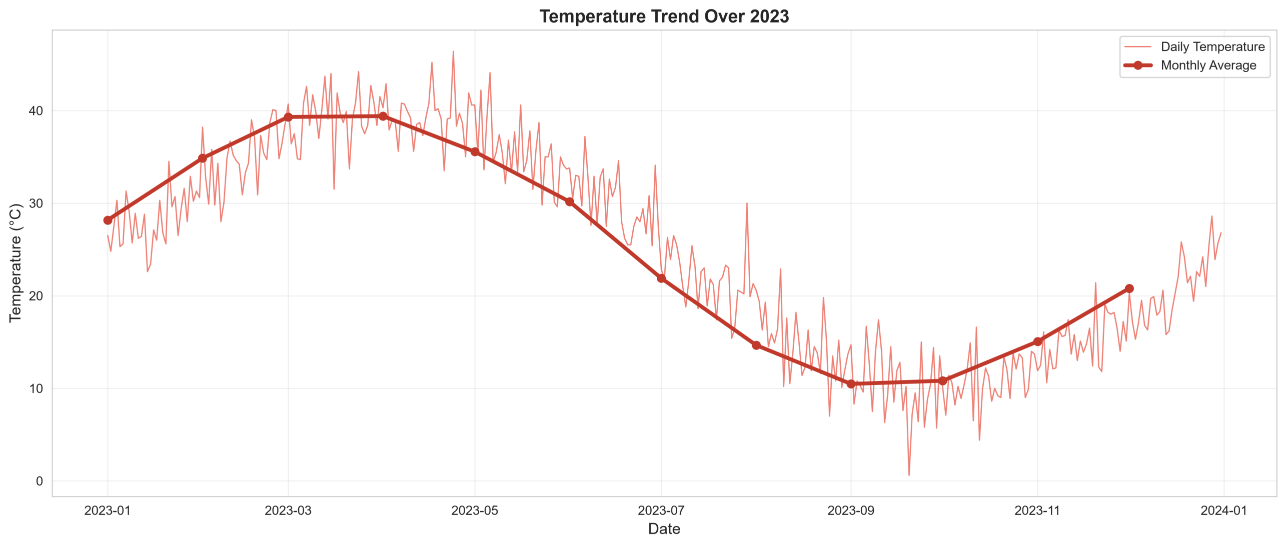
WEATHER DISTRIBUTION:
- Hot Days: 119 (32.6%)
- Pleasant Days: 106 (29.0%)
- Rainy Days: 77 (21.1%)
- Cold Days: 63 (17.3%)

KEY INSIGHT:
The dataset shows strong seasonal variation in rainfall, with substantial rainfall concentrated during June-September. Temperature also varies considerably across the defined seasons. These observed patterns can support planning for water management, agriculture, infrastructure, and weather monitoring.

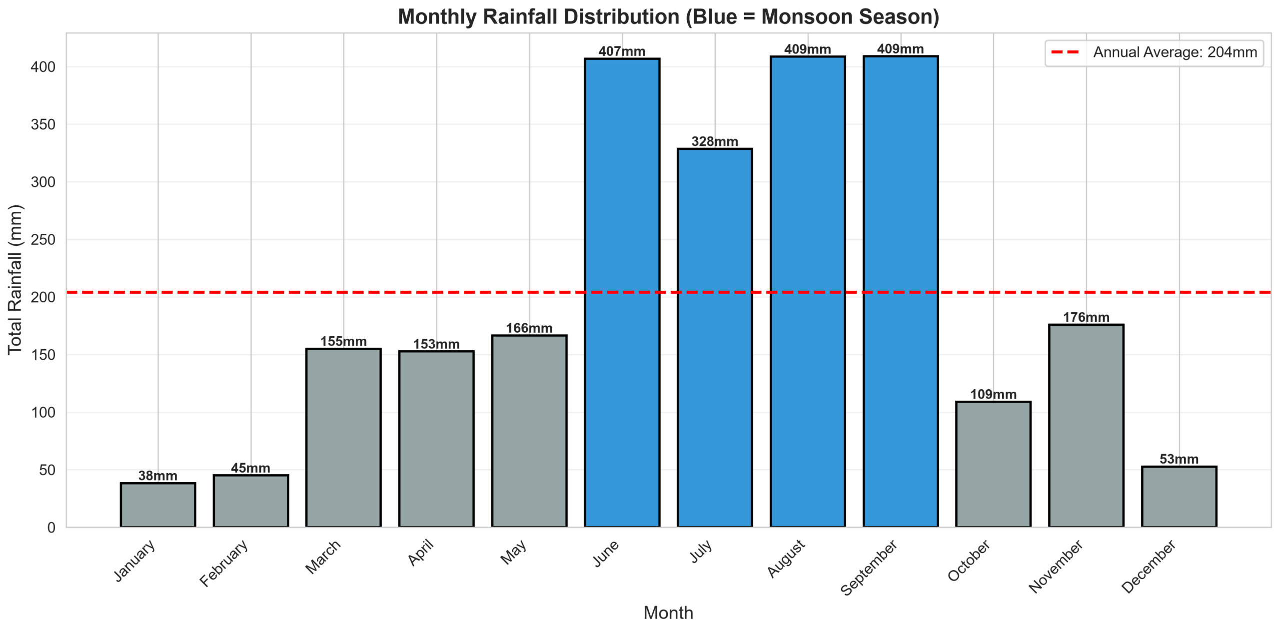
The analysis describes patterns within the supplied one-year dataset and does not independently establish a formal long-term climate classification.

Temperature & Rainfall Trends

Temperature Trend Over 2023

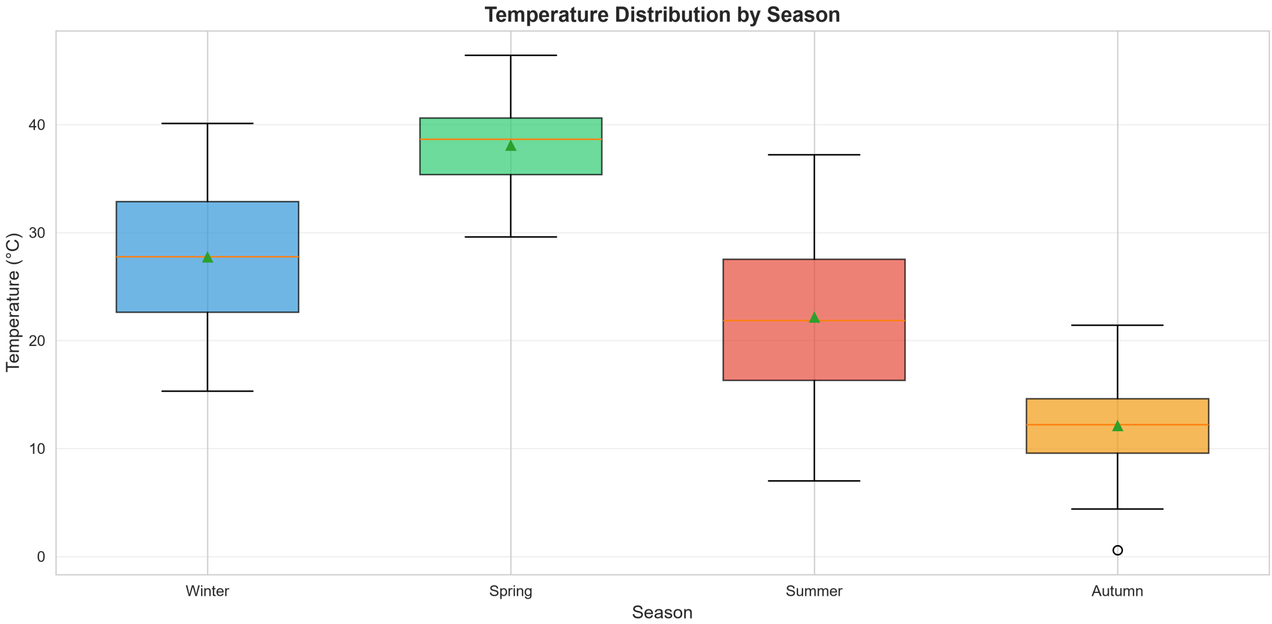


Monthly Rainfall Distribution

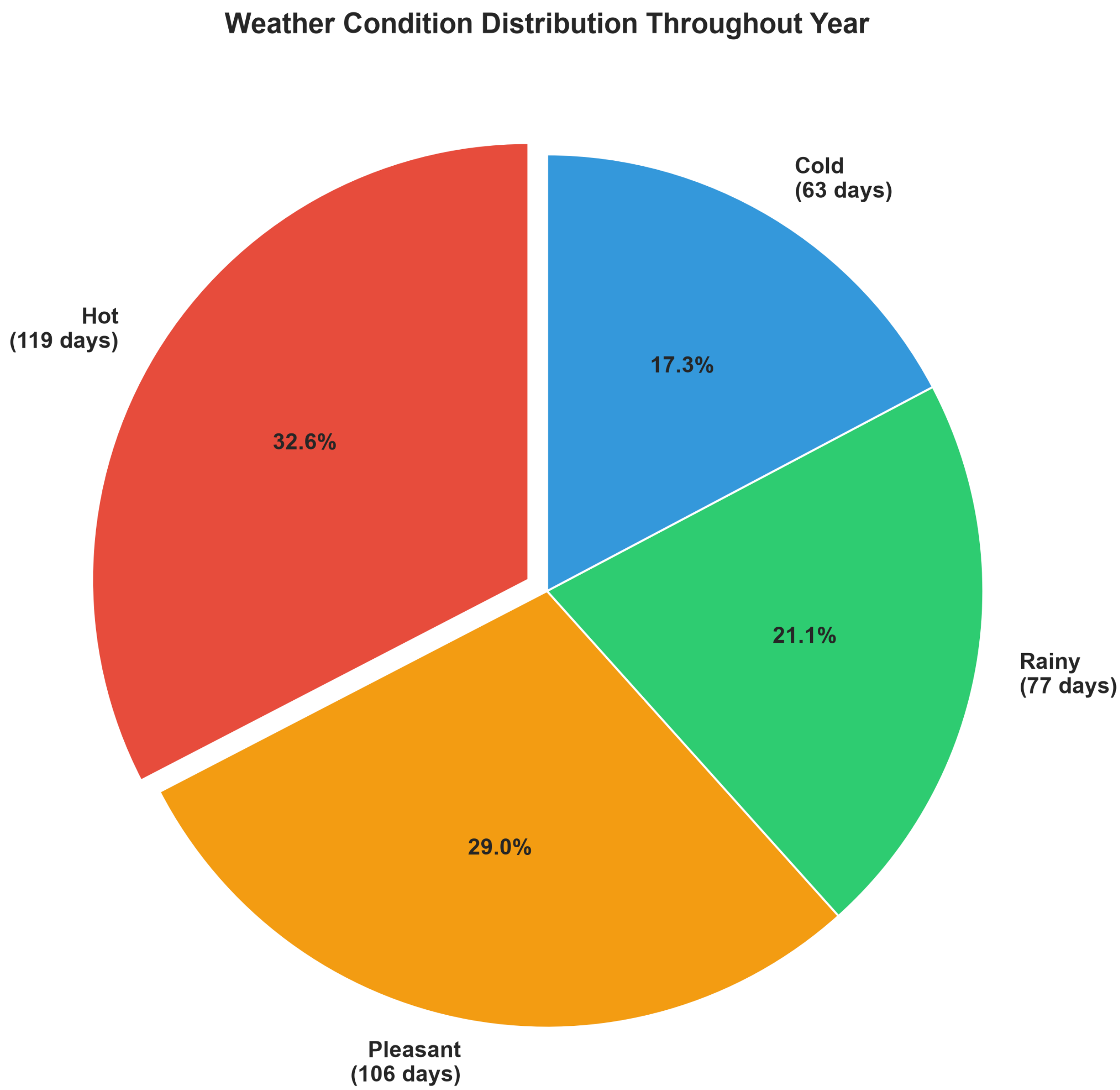


Seasonal & Condition Analysis

Seasonal Temperature Distribution



Weather Condition Distribution



Weather Condition Distribution Throughout Year
Cold
(63 days)
17.3%
Hot
(119 days)
32.6%
21.1%
Rainy
(77 days)
29.0%
Pleasant
(106 days)

Detailed Findings

1. STRONG SEASONAL RAINFALL PATTERN
The dataset shows substantial rainfall concentration during June-September. 77 heavy-rain days account for approximately 65% of annual rainfall, indicating strong seasonal concentration in precipitation. This pattern is consistent with a monsoon-like seasonal rainfall cycle in the supplied data.

2. SIGNIFICANT TEMPERATURE VARIATION BY DEFINED SEASONS
The defined seasonal groups show substantial differences in average temperature. Spring has the highest average temperature at approximately 38°C, while autumn has the lowest at approximately 12°C, producing a difference of about 26°C between the seasonal averages.

3. RAINFALL OCCURS THROUGHOUT THE YEAR
The dataset contains 284 days classified as rainy, representing approximately 78% of the year. However, rainfall intensity is not evenly distributed across these days, with heavy rainfall concentrated mainly during June-September.

4. WEAK LINEAR CORRELATIONS BETWEEN WEATHER VARIABLES
Temperature, humidity, rainfall, wind speed, and pressure show relatively weak pairwise linear correlations, generally below 0.15 in magnitude. This indicates that simple linear relationships between individual weather variables are limited in this dataset.

5. EXTREME OBSERVATIONS SHOW SEASONAL CONCENTRATION
Hot observations above 31.5°C are concentrated mainly during March-May and September-October, while cold observations below 4.4°C occur mainly during November-February. Heavy rainfall above 10mm is concentrated primarily during June-September.

6. RAINFALL INTENSITY DISTRIBUTION
The 77 heavy-rain days account for approximately 1,583mm of rainfall. Average daily rainfall is approximately 6.71mm, indicating a combination of lower-intensity rainfall days and a smaller number of heavier rainfall events.

Strategic Recommendations

AGRICULTURAL PLANNING:
- Use the observed rainfall calendar to support crop and irrigation planning.
- Consider the June-September period when planning activities sensitive to heavy rainfall.
- Evaluate crop and soil requirements against observed rainfall patterns before making cultivation decisions.
- Use rainfall monitoring to support irrigation scheduling during lower-rainfall periods.

WATER RESOURCE MANAGEMENT:
- The concentration of approximately 65% of annual rainfall within 77 heavy-rain days highlights the potential value of rainwater storage and runoff management.
- Assess storage capacity using historical rainfall concentration as a planning input.
- Monitor rainfall intensity during June-September for water-management and flood-preparedness decisions.

INFRASTRUCTURE & CONSTRUCTION:
- Consider the observed maximum daily rainfall of 86.4mm when evaluating drainage requirements.
- Incorporate seasonal rainfall patterns into construction scheduling and site-management plans.
- Use observed temperature extremes when evaluating material and environmental requirements.

ENVIRONMENTAL MONITORING:
- Increase rainfall monitoring during June-September, when heavy rainfall is most concentrated.
- Monitor temperature extremes during periods where hot and cold observations are most frequent.
- Track rainfall intensity and accumulation to support flood-risk assessment.
- Compare future observations with this 2023 baseline to identify changes in seasonal patterns.

FORECASTING & WARNING SYSTEMS:
- The weak pairwise linear correlations indicate that individual weather variables should not be treated as sufficient predictors on their own.
- Future forecasting work could evaluate multi-variable statistical or machine-learning models.
- Historical rainfall and temperature patterns can be used as baseline information for monitoring future observations.

Data Interpretation & Limitations

This analysis is based on 365 days of observations from a single weather station and is intended primarily to describe patterns within the supplied dataset.

1. ONE-YEAR OBSERVATION PERIOD
A single year is sufficient for identifying observed seasonal patterns but is not sufficient by itself to establish long-term climate characteristics or climate trends.

2. SINGLE-STATION DATA
The analysis represents the supplied weather station and should not automatically be generalized to an entire city, region, or climate zone.

3. CLIMATE CLASSIFICATION
The observed rainfall and temperature patterns are consistent with a monsoon-like seasonal pattern, but a formal climate classification would normally require longer-term observations and additional climatological evidence.

4. CORRELATION DOES NOT ESTABLISH CAUSATION
The reported correlations describe statistical relationships between variables in the dataset. They do not establish causal relationships.

5. FORECASTING LIMITATION
Historical patterns can provide useful baselines, but they do not guarantee future weather conditions. More advanced forecasting would require additional historical observations and potentially external meteorological variables.

6. PLANNING RECOMMENDATIONS
The recommendations in this report are analytical planning considerations derived from observed patterns. They should be combined with local environmental, agricultural, engineering, and meteorological expertise before operational decisions are made.

Conclusion

The weather dataset shows substantial seasonal variation in both temperature and rainfall. Approximately 65% of annual rainfall is concentrated within 77 heavy-rain days, primarily during June-September, while the defined seasonal groups show a large difference in average temperature.

The analysis also identifies seasonal concentration of hot, cold, and heavy-rain observations. At the same time, the relatively weak pairwise correlations between weather variables indicate that simple linear relationships provide limited explanatory power within this dataset.

These findings provide useful baseline information for agricultural planning, water-resource management, infrastructure preparation, environmental monitoring, and future forecasting analysis.

However, the results should be interpreted as observations from a single weather station over one year. The dataset demonstrates a monsoon-like seasonal rainfall pattern, but it does not by itself establish a formal long-term climate classification or guarantee future weather behavior.

Overall, the analysis provides a structured quantitative view of the supplied 2023 weather observations while highlighting both their practical planning value and their analytical limitations.